

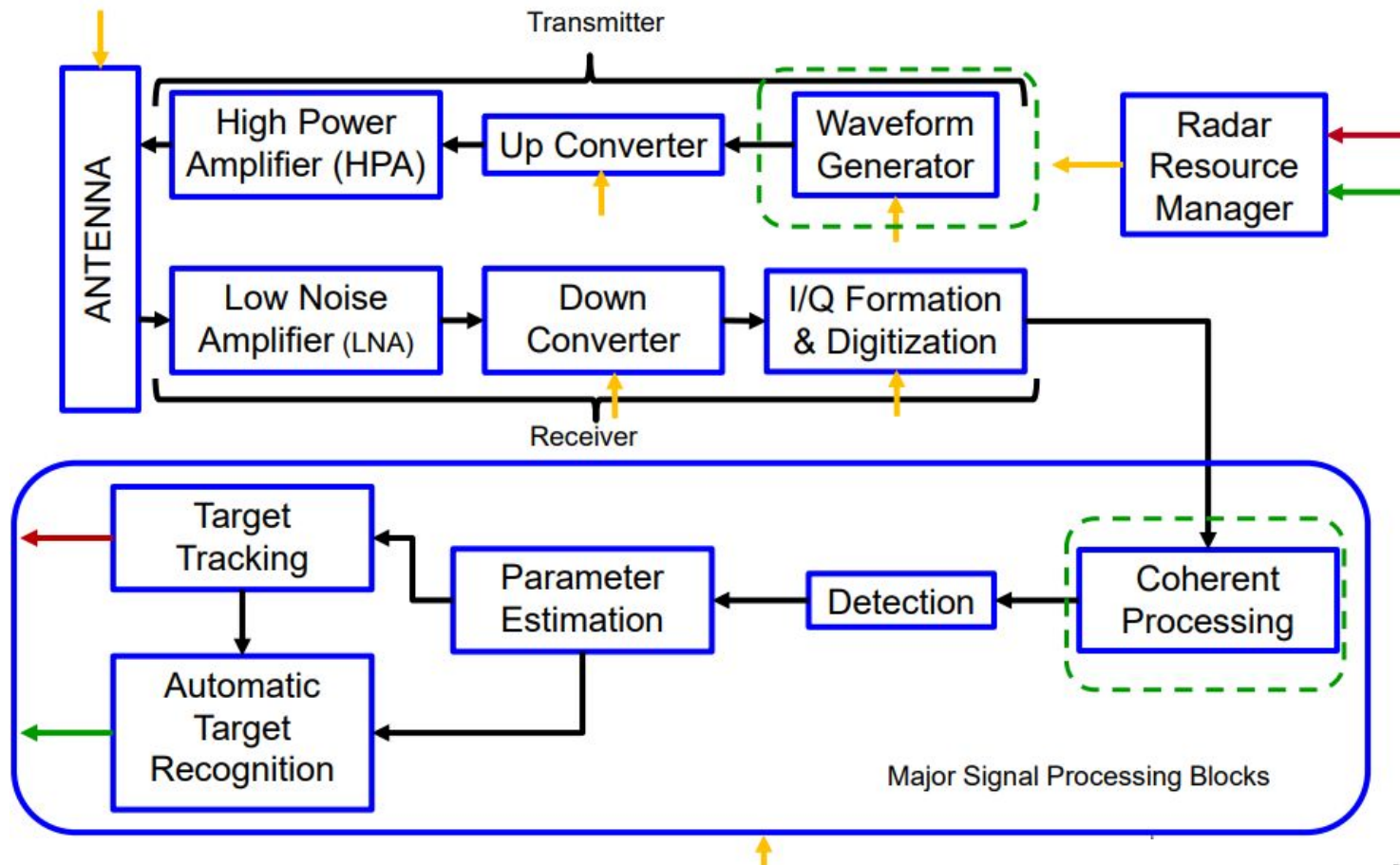
FRSP Lecture 4

The Matched Filter and Range Resolution

EE417 / EE517
CUA Spring 2024

Tuesdays 7:00-9:30pm
Instructor: Prof. Kevin Russo

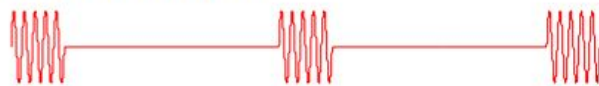
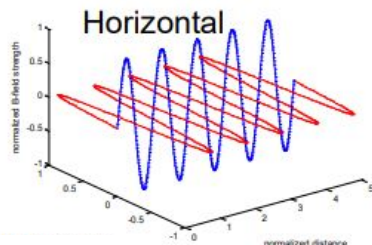
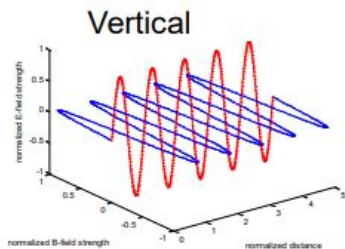
Generic Radar System Block Diagram



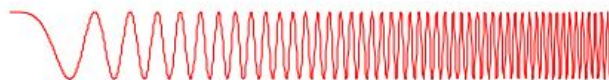
Waveforms

- The term “waveform” refers to the modulation applied by the radar to the EM wave
- General classes of modulation

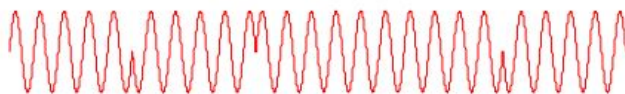
- Amplitude
- Frequency
- Phase
- Polarization



“ON-OFF” Amplitude Modulation



Frequency Modulation



Phase Modulation

Examples

$$\bar{x}(t) = a(t) \sin(\Omega t + \theta(t)) \quad \text{Real signal with carrier}$$

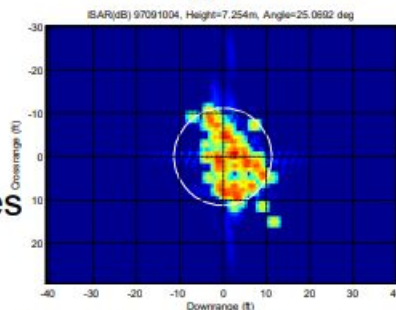
$$\bar{x}(t) = a(t) \exp(j[\Omega t + \theta(t)]) \quad \text{Complex signal with carrier}$$

$$x(t) = a(t) \exp(j\theta(t)) \quad \text{Complex signal without carrier}$$

Waveforms Extract “Target” Information

- Radar – Radio Detection and Ranging
- Waveforms are designed to
 - Facilitate the extraction of information about a target/object
 - Information regarding a target/object
 - Presence or absence - Detection
 - Range to object
 - Speed of object
 - Physical size of object
 - Radar cross section (reflective properties)
 - Range, Doppler, and cross-range profiles/images
 - Polarization response
 - Object track
 - Maximize the desired signal return while suppressing the contribution from interference
 - Clutter (e.g.,
 - Scattering from terrain
 - Scattering from precipitation
 - Terrain Range/Doppler ambiguities
 - Thermal Noise
 - Electronic Attack (Electronic Protection – PRF, Freq, Modulation, etc.)

SAR Image of a Tank



Radar Frequency Bands

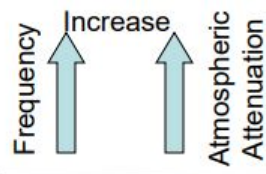
Designator	Frequency		Units	Wavelength (m)	
	Lower	Upper			
VHF	50	300	MHz	6.000	1.000
UHF	300	1000	MHz	1.000	0.300
L	1	2	GHz	0.300	0.150
S	2	4	GHz	0.150	0.075
C	4	8	GHz	0.075	0.038
X	8	12	GHz	0.038	0.025
Ku	12	18	GHz	0.025	0.017
K	18	27	GHz	0.017	0.011
Ka	27	40	GHz	0.011	0.008
V	40	75	GHz	0.008	0.004
W	75	100	GHz	0.004	0.003



Some Factors

$$\theta_{BW} \approx \frac{\lambda}{D}$$

Antenna Beamwidth



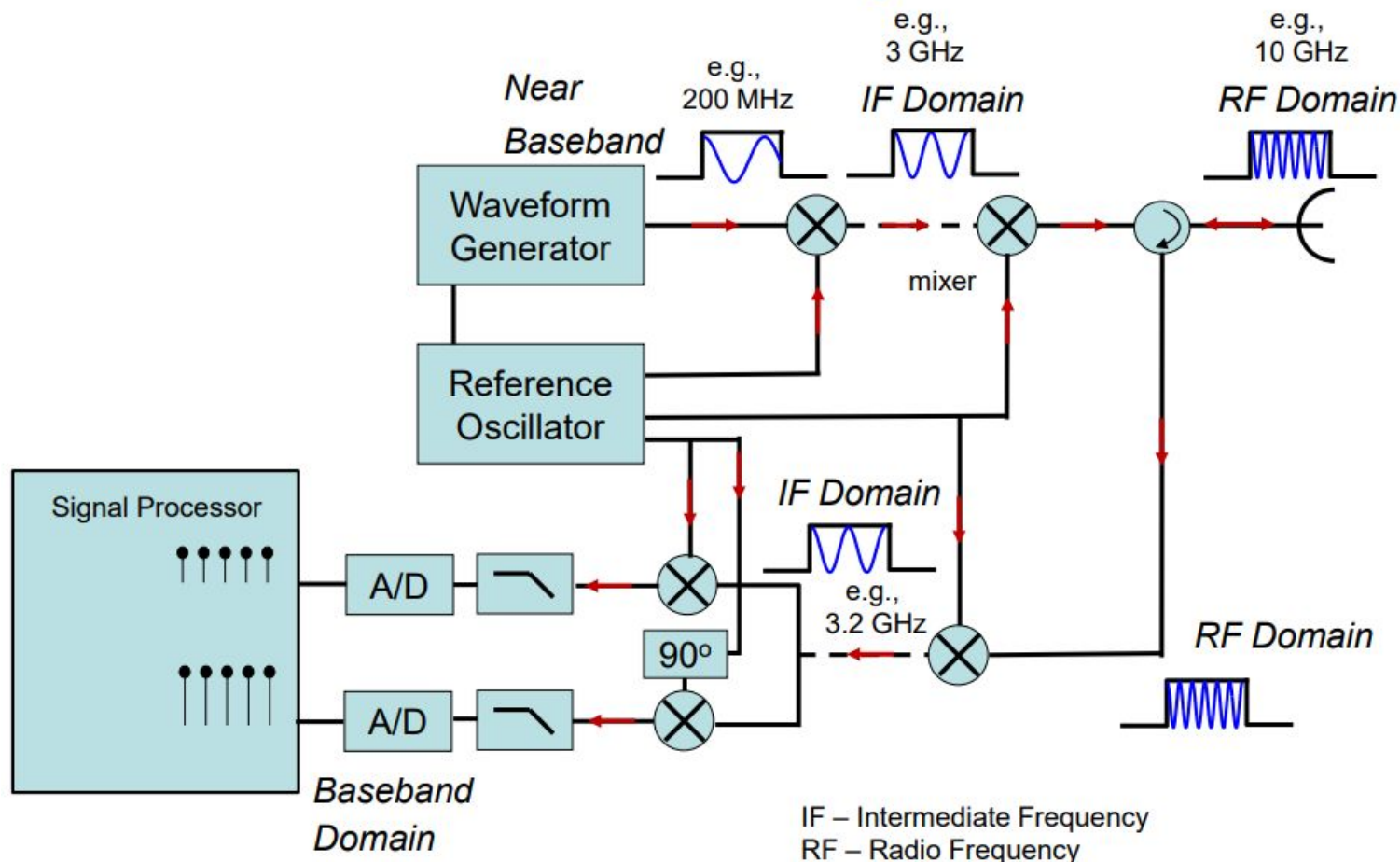
$$f_{Doppler} = -\frac{2v}{\lambda}$$

Doppler Shift

Designator	Applications
VHF	Very long-range surveillance
UHF	Very long-range surveillance
L	Long-range surveillance, enroute traffic control
S	Moderate-range surveillance, terminal traffic control, long-range weather
C	Long-range tracking
X	Short-range tracking, missile guidance, mapping, marine radar, airborne intercept, airborne weather
Ku	High resolution mapping
K	(H ₂ O absorption); automotive radars
Ka	Very high resolution mapping
V	Experimental
W	Automotive radars; Experimental - real beam landing aids

MMW
Up to
300 GHz

Generic Radar System



Signal-To-Noise Ratio

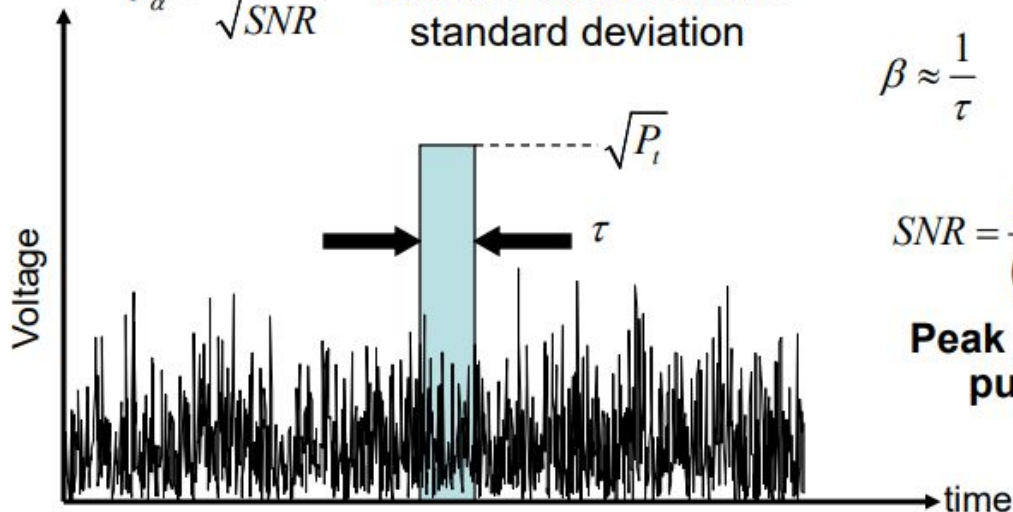
- Target detection and measurement accuracy are a functions of signal-to-noise ratio (*SNR*)
- Desire is to maximize *SNR*

$$P_D = \frac{1}{P_{FA}^{1+SNR}}$$

Swerling I Target

$$\sigma_\alpha = \frac{\delta\alpha}{\sqrt{SNR}}$$

General measurement standard deviation



$$SNR = \frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 R^4 k T_s \beta}$$

Basic Radar Range Equation

$$\beta \approx \frac{1}{\tau}$$

For an un-modulated pulse, bandwidth is inversely proportional to pulse width

$$SNR = \frac{P_t \tau G_t G_r \lambda^2 \sigma}{(4\pi)^3 R^4 k T_s}$$

Energy Form of Radar Range Equation

Peak power limitations drive longer pulse lengths to achieve SNR requirements

The Matched Filter

Matched filter

- maximizes SNR at a point in time corresponding to the time-delay to the target

Transmit waveform $x(t) \quad 0 \leq t \leq \tau$

Matched filter $h(t) = K \cdot x^*(-t) \quad -\tau \leq t \leq 0 \quad H(\omega) = KX^*(\omega)$

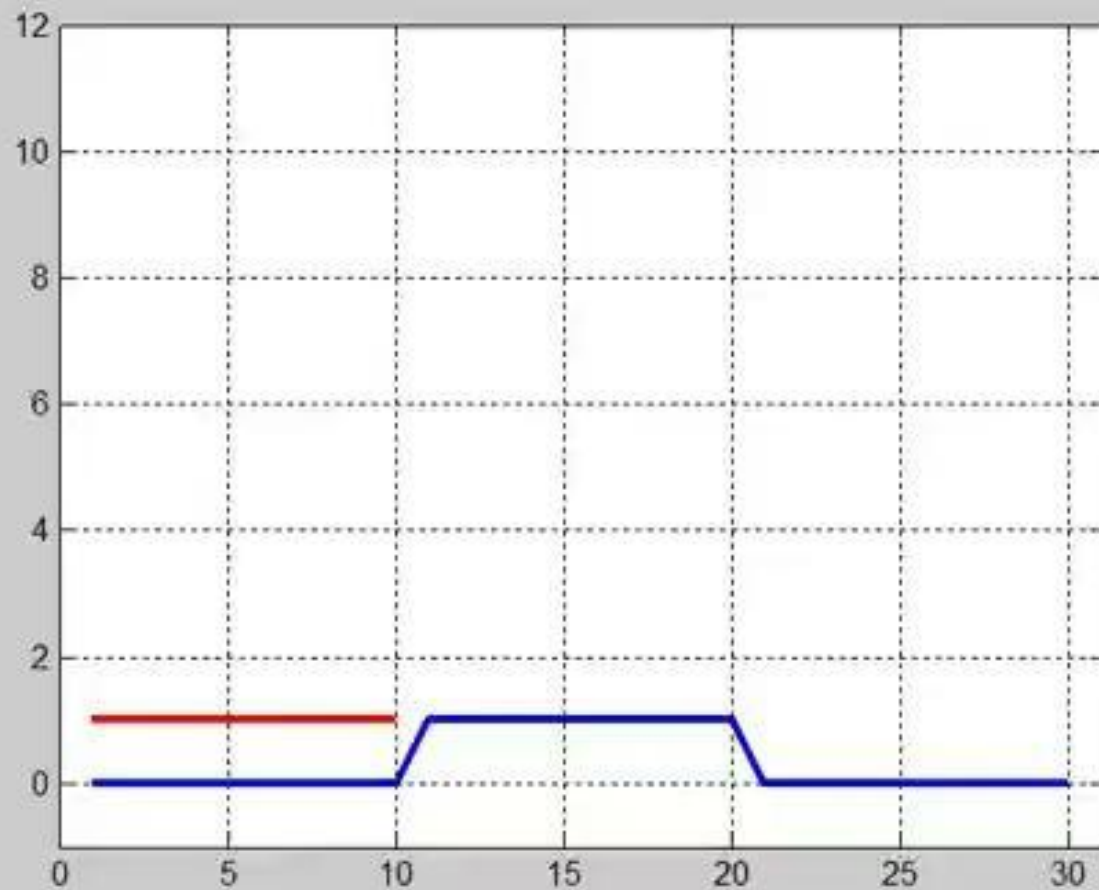
where K is an arbitrary constant

The filter operation is defined in terms of a convolution integral

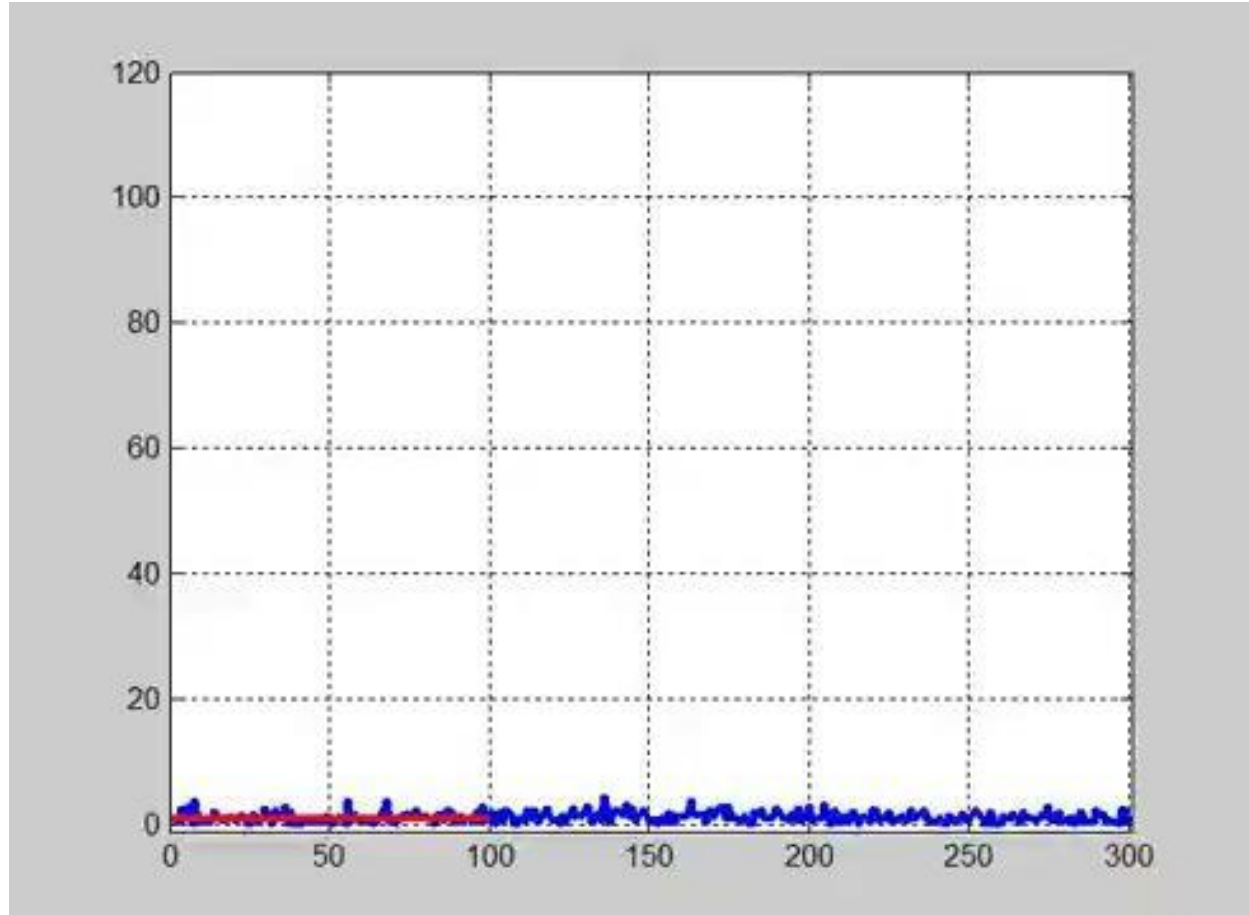
$$y(t) = \int_{-\infty}^{\infty} x(s)h(t-s)ds = K \int_{-\infty}^{\infty} x(s)x^*(s-t)ds$$

s is dummy
variable of
integration

Equivalent to correlating $x(t)$ with $Kx(t)$

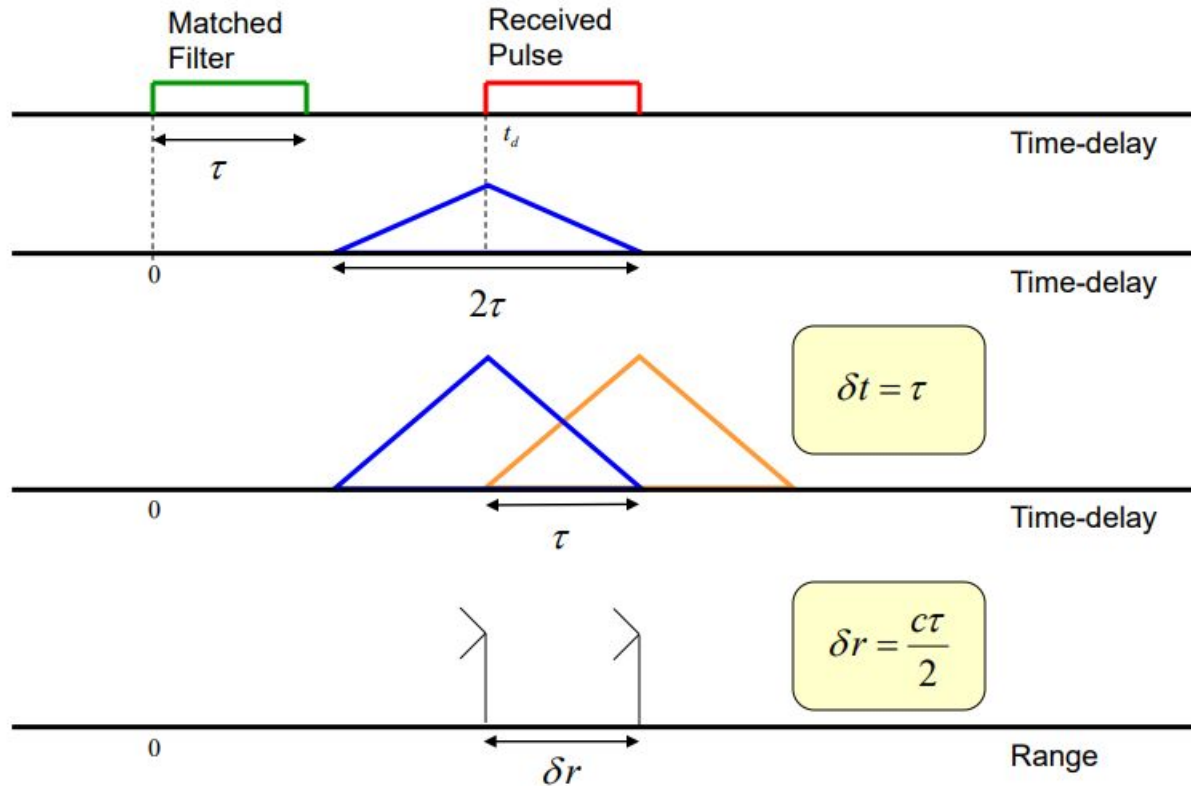


mf_movie_noise.mpg

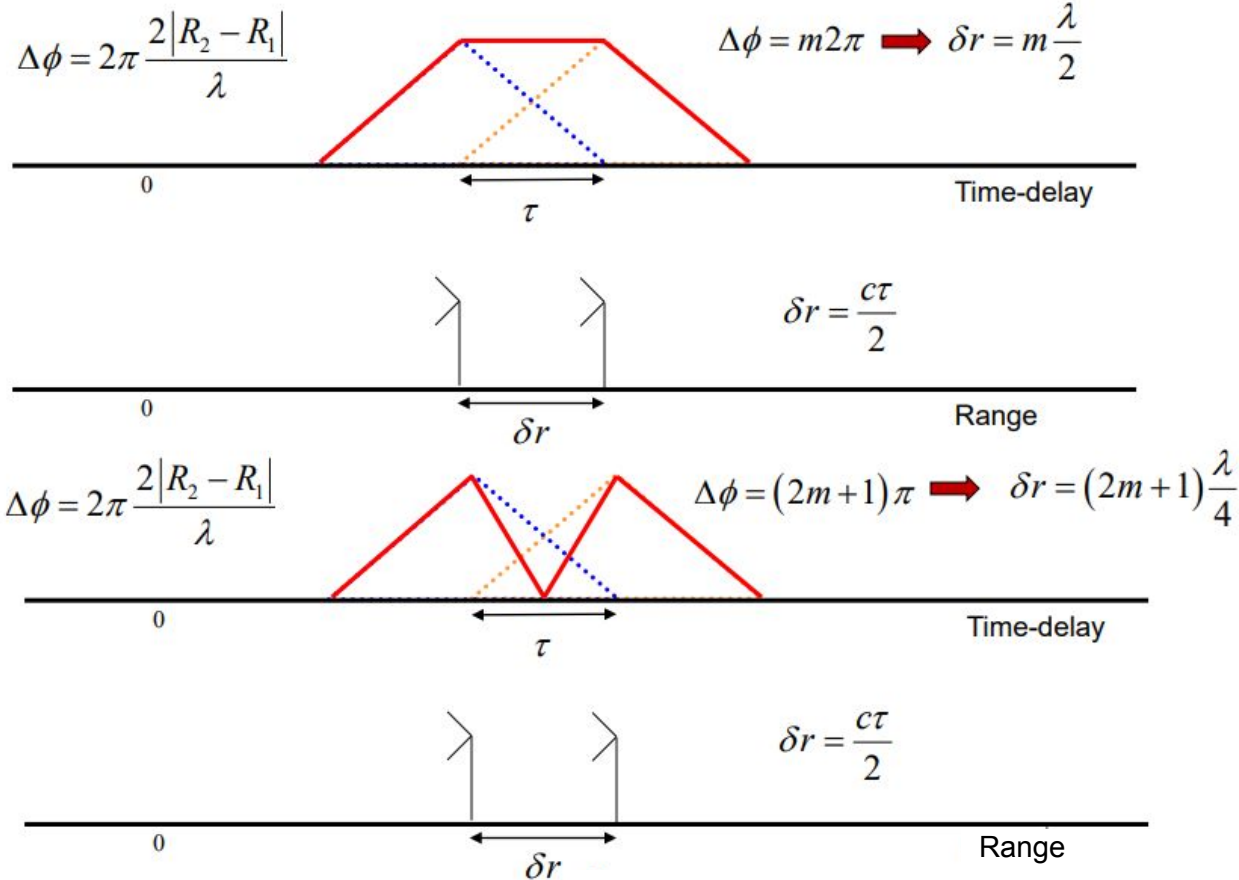


Range Resolution & The Matched Filter

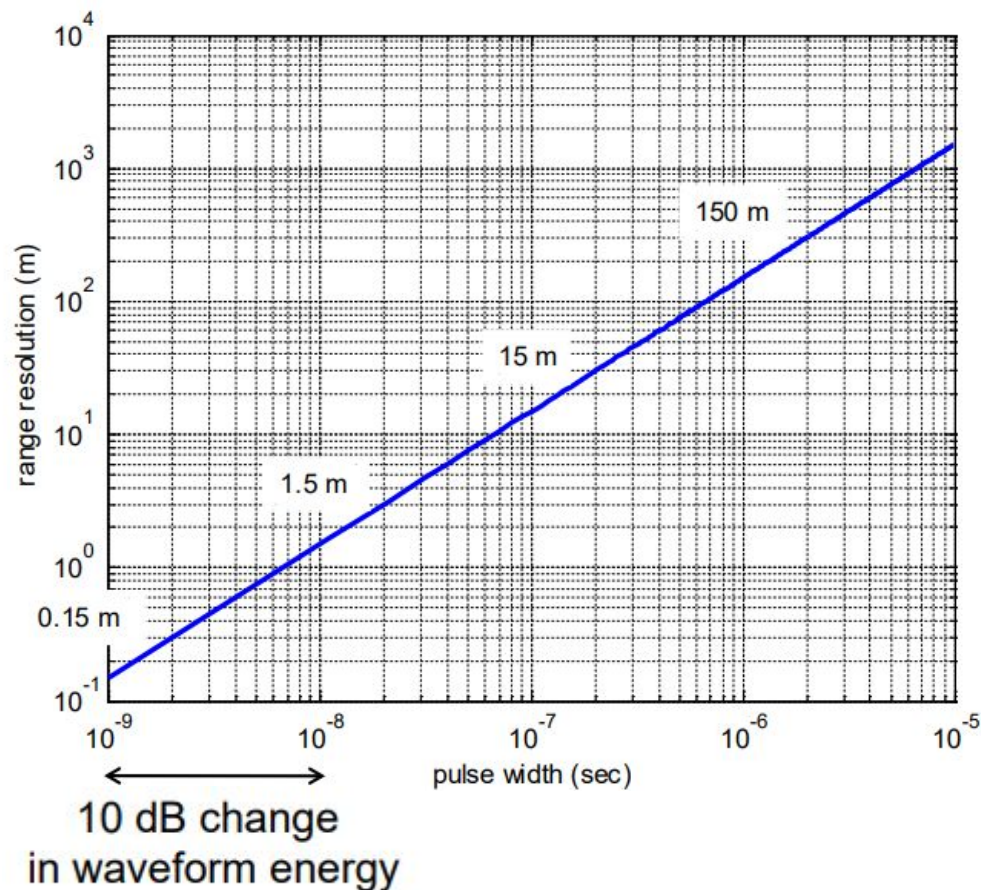
Rayleigh Resolution – separation between the peak and first null

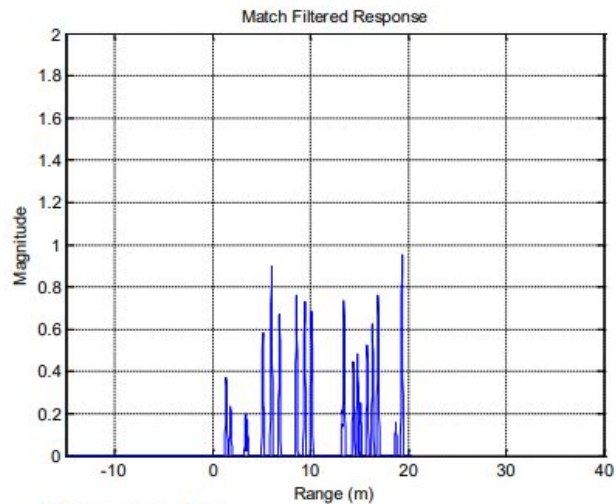


Phase and Range Resolution



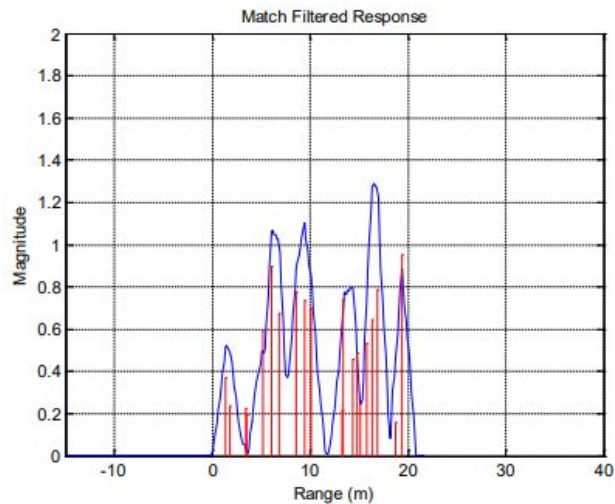
Pulse Width and Range Resolution (Simple Pulse)





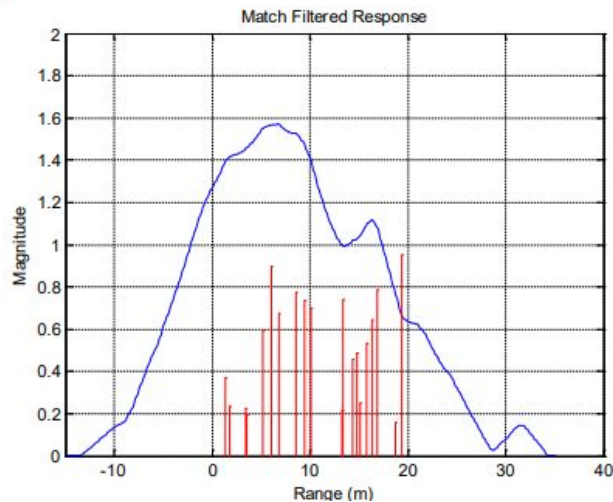
1 nsec pulse

$\delta r = 0.15$ m



10 nsec pulse

$\delta r = 1.5$ m



100 nsec pulse

$\delta r = 15$ m